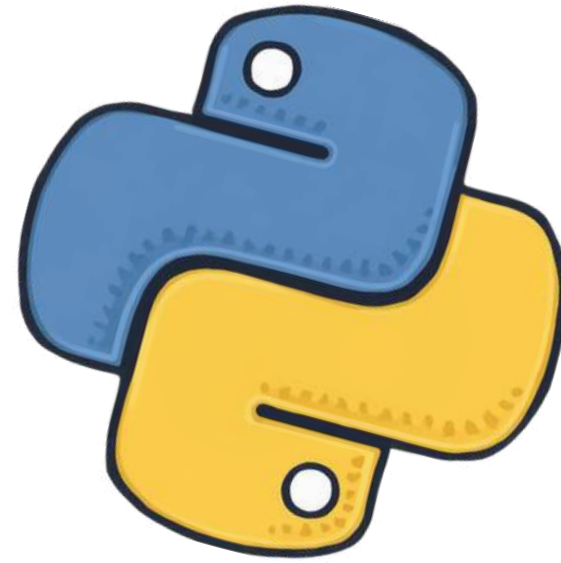




TUPLES & SETS

PYTHON'S SPECIAL COLLECTIONS

WHAT ARE COLLECTIONS IN PYTHON?

Collections are groups of items stored together in Python! Think of it like organizing your things — you can list your toys, your school books, or your friends' names all in one place.

Python gives us different types of collections for different jobs. Today we'll meet two special ones: tuples and sets.

Lists → Tuples → Sets



WHAT IS A TUPLE?

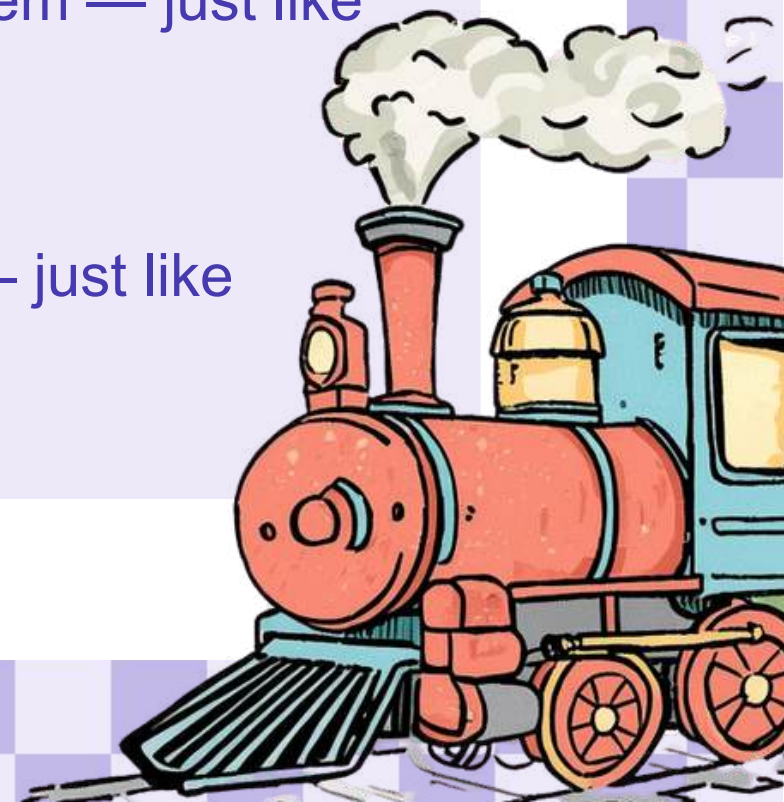
A tuple is a collection of items that stays in order and cannot be changed. Think of it like a train with fixed coaches — once the coaches are set, they stay in the same order!

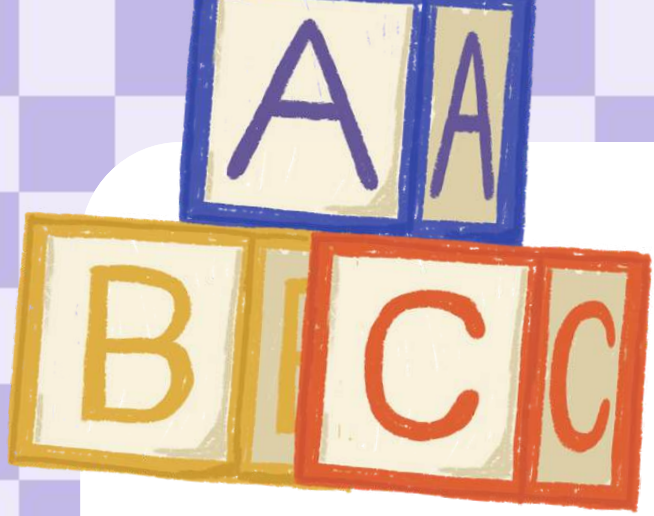


Real-life analogy: The Rajdhani Express

The coaches S1, S2, S3, AC1, AC2 are always in the same order. You cannot swap or rearrange them — just like a tuple! Once the train is set, the order stays the same forever.

Another example: your Aadhaar card has your name, date of birth, and number that never change — just like tuple data!





CREATING & ACCESSING TUPLES

Create a tuple with parentheses (), items separated by commas:

```
train_coaches = ("S1", "S2", "S3", "AC1", "AC2")  
print(train_coaches)
```

Use index numbers to get items — index starts at 0, not 1!

```
print(train_coaches[0])    # S1  
print(train_coaches[3])    # AC1  
print(len(train_coaches))  # 5
```

Use ()

Items separated by commas

Order stays the same

TUPLE PROPERTIES

Ordered

Immutable

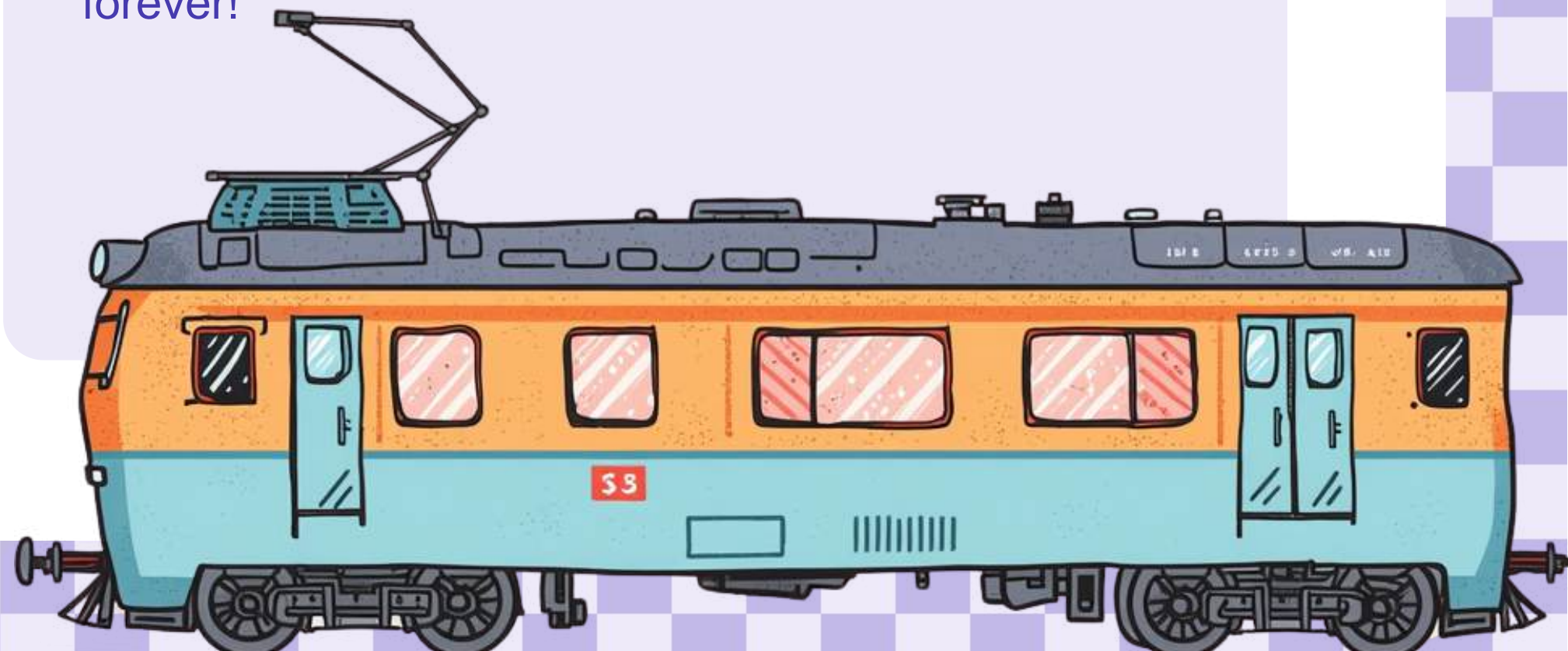
Duplicates OK

Mixed Types

— What happens if you change a tuple?

```
train_coaches[0] = "B1"  
# TypeError!
```

You cannot change a tuple once it is made. Tuples are immutable, which means their items stay fixed forever!



WHAT IS A SET?

A set is a collection of items that has no order and no duplicates. Think of it like a dabba (tiffin box) with unique dals — you can have rajma, chana, and moong, but no dal appears twice!

Real-life analogy: The Dabba (Tiffin Box)

Imagine your school dabba has rajma, chana, moong, and you accidentally add rajma again. In a set, the duplicate rajma disappears automatically!

Another example: during roll call, each student's name appears only once in the register — no repeats allowed! That's exactly how sets work.



CREATING SETS & THEIR PROPERTIES

Use curly braces {} to create a set:

```
dals = {"rajma", "chana", "moong", "rajma"}  
print(dals)  
# {'rajma', 'chana', 'moong'}
```

Items go in curly braces, separated by commas — duplicates are removed automatically!

Unordered

Mutable

No Duplicates

No Indexing



CHANGING SETS — ADD, REMOVE & CHECK

Sets can change after creation — use `.add()` and `.discard()`:

```
student_names = {"Priya", "Arjun", "Sneha"}
student_names.add("Kiran")
student_names.discard("Sneha")
print(student_names)
```

❌ `student_names[0]`
TypeError!
Sets are unordered —
no indexing allowed.

Check if an item exists using the `'in'` keyword:

```
print("biryani" in my_foods)    # True
print("pizza" in my_foods)     # False
```



TUPLE VS SET VS LIST

Feature	List	Tuple	Set
Symbol	[]	()	{}
Ordered?	✓	✓	✗
Can Change?	✓	✗	✓
Duplicates?	✓	✓	✗
Indexing?	✓	✓	✗

Same idea, three different rules! Choose the collection that fits your data.



WHEN TO USE TUPLES VS SETS?

Use a Tuple when data should stay fixed:

Days of the week, date of birth

Fixed IPL teams for the season

Coordinates of your school

Indian states in order

Use a Set when you want unique items only:

Unique states visited

Languages spoken in class

Canteen items

Sports played at school

CODING ACTIVITY — MY INDIAN DATA BOX

Time to code! Create your own tuple with Indian states and a set with your favorite Indian foods, then try the challenges below.

```
my_states = ("Maharashtra", "Kerala",  
            "Rajasthan", "Punjab", "Goa")  
print(my_states[0])    # Maharashtra  
my_states[0] = "Gujarat" # TypeError!
```

```
my_foods = {"idli", "dosa", "paratha",  
            "biryani", "idli"}  
my_foods.add("pani puri")  
print("biryani" in my_foods) # True
```

Notice: the tuple keeps its fixed order and rejects changes, while the set drops the repeated "idli" and grows with .add()!



SUMMARY & KEY VOCABULARY

Tuple: a fixed-order collection that cannot be changed, like train coaches.

Set: a collection of unique items with no fixed order, like dals in a dabba.

Immutable: cannot be changed once created.

Index: an item's position, starting from 0.

Tuples (): ordered, unchangeable, duplicates allowed — like a train with fixed coaches!

Sets { }: unordered, changeable, no duplicates — like a dabba with unique dals!

Thank You! Questions?

